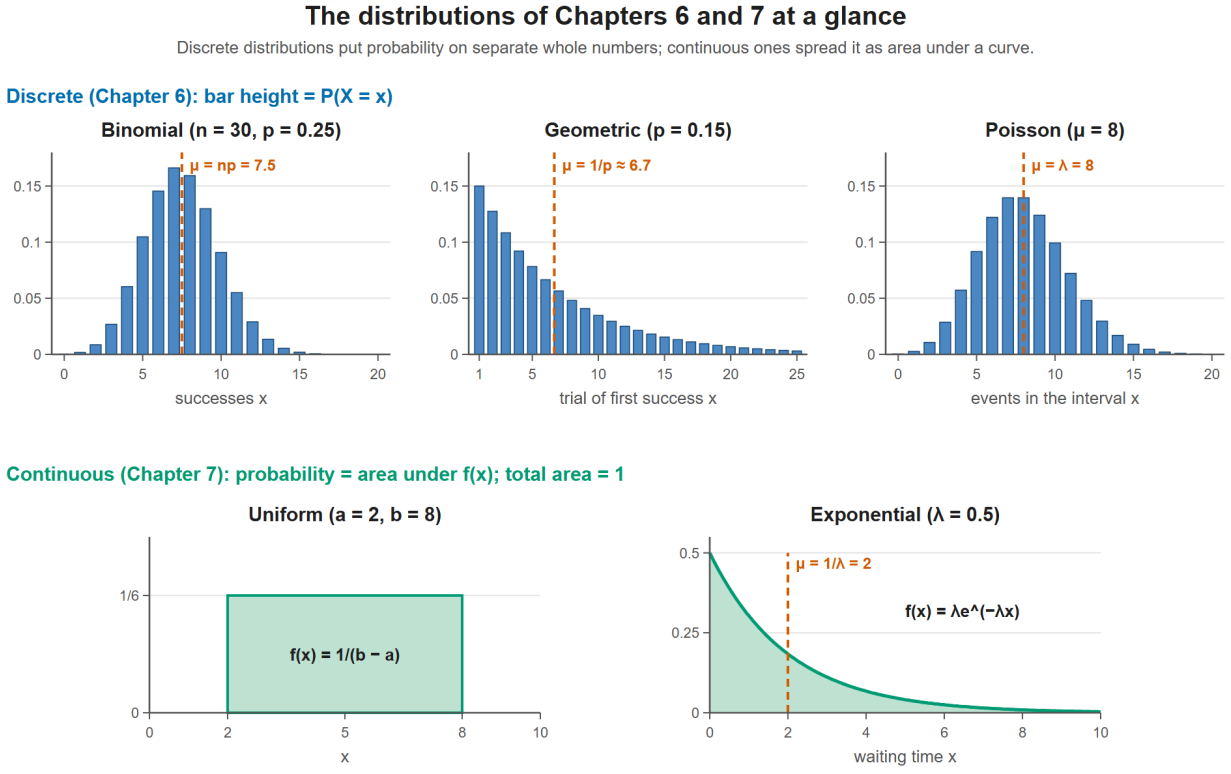
Lesson 6.1 - Probability Distribution Formulas

> **Video:** Probability Distribution Formulas by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

A reference sheet for the distributions in Chapters 6 and 7. Throughout, p = probability of success and $q = 1 - p$ = probability of failure.



What each distribution looks like. The three discrete ones (top) give probability as the height of a bar at a whole number. The two continuous ones (bottom) give probability as the area under the curve, and the total area is always 1. The dashed lines mark each mean.

---

1. Binomial distribution

X = number of successes in n independent trials.

$$P(X = x) = {}_nC_x \, p^x q^{n-x}, \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Mean	Standard deviation
$\mu = np$	$\sigma = \sqrt{npq}$

---

2. Geometric distribution

X = the trial on which the **first success** happens.

$$P(X = n) = q^{n-1}p$$

Why: for the first success to be on trial 4, trials 1-3 must fail ($q \cdot q \cdot q$) and trial 4 must succeed (p), so $P(X = 4) = q^3p$.

Probability	Formula
$P(X > n)$	q^n
$P(X \geq n)$	q^{n-1}
$P(X \leq n)$	$1 - q^n$
$P(X < n)$	$1 - q^{n-1}$

Mean	Variance	Standard deviation
$\mu = \frac{1}{p}$	$\sigma^2 = \frac{1}{p}\left(\frac{1}{p} - 1\right) = \mu(\mu - 1) = \frac{q}{p^2}$	$\sigma = \frac{\sqrt{1-p}}{p}$

---

3. Poisson distribution

X = number of events in a fixed interval, where the average number of events is $\mu = \lambda$.

$$P(X = n) = \frac{\mu^n e^{-\mu}}{n!} = \frac{\lambda^n e^{-\lambda}}{n!}$$

Probability	Formula
$P(X \leq n)$	$e^{-\mu}\sum_{x=0}^{n}\frac{\mu^x}{x!}$
$P(X > n)$	$1 - e^{-\mu}\sum_{x=0}^{n}\frac{\mu^x}{x!}$

Mean	Variance	Standard deviation
$\mu = \lambda$ ($= np$ when approximating a binomial)	$\sigma^2 = \lambda$	$\sigma = \sqrt{\lambda}$

> In the video the Poisson denominator is spoken as "n squared". The correct denominator is $n!$ (n factorial), as in the other form written on screen.

---

4. Uniform distribution (continuous)

X is equally likely to be anywhere between a and b. The graph is a **rectangle** of width $b-a$, and its total area must be 1:

$$\text{area} = \text{base} \times \text{height} = (b-a)\cdot f(x) = 1 \;\Rightarrow\; f(x) = \frac{1}{b-a}$$

Probability = partial width ÷ total width:

$$P(c \le X \le d) = \frac{d-c}{b-a}$$

For example, $P(a \le X \le c) = \dfrac{c-a}{b-a}$ and $P(c \le X \le b) = \dfrac{b-c}{b-a}$.

Mean	Standard deviation
$\mu = \dfrac{a+b}{2}$	$\sigma = \dfrac{b-a}{\sqrt{12}}$

---

5. Exponential distribution (continuous)

Waiting times. λ = rate parameter, and $\mu = 1/\lambda$.

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$$

Probability	Formula	From calculus
$P(X \le n)$ (left area)	$1 - e^{-\lambda n}$	$\displaystyle\int_0^n \lambda e^{-\lambda x}\,dx$
$P(X \ge n)$ (right area)	$e^{-\lambda n}$	$\displaystyle\int_n^\infty \lambda e^{-\lambda x}\,dx$

Mean	Variance	Rate
$\mu = \dfrac{1}{\lambda}$	$\sigma^2 = \dfrac{1}{\lambda^2}$	$\lambda = \dfrac{1}{\mu}$

In general, a probability for a continuous distribution is the **area under** $f(x)$: $P(a \le X \le b) = \int_a^b f(x)\,dx$.

Where each one is used

Distribution	Typical question
Binomial	"Exactly / at least x successes in n trials" (Lesson 6.2)
Geometric	"First success on trial n" (Lesson 6.3)
Poisson	"Number of events in a time or space interval" (Lesson 6.4)
Uniform	"Equally likely anywhere between a and b" (Lesson 7.3)
Exponential	"Waiting time until the next event" (Lesson 7.4)